Cryptography and Embedded System Security
CRAESS_I

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Timetable

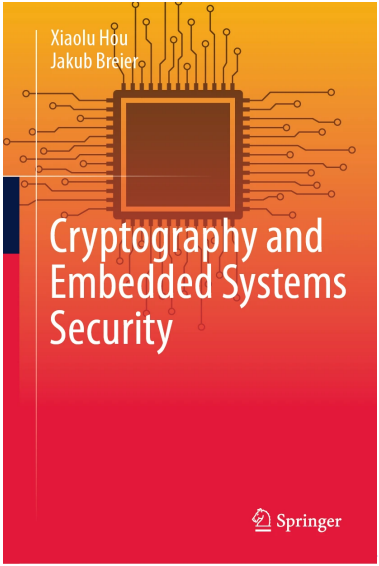
- Lectures (prednášky)
  - Friday 10:00–11:50
- Tutorials (cvičenia)
  - Friday after the lecture
  - 12:00–13:50
- Consultations (konzultácie)
  - By appointment: `xiaolu.hou@stuba.sk`
  - Office 4.03

Grading

- 60 marks – six assignments
  - Assignments with questions
    - Solutions must be written in LaTeX; submit the PDF in AIS
  - Programming assignments
    - Submission of code in AIS
    - Individual presentation of implementation details
    - Assignments 4 and 5 also require short answers for a few questions to be submitted in PDF in AIS
  - Grade 0 for late submission
- 10 extra marks
  - Find mistakes in the textbook
- 40 marks – final exam
  - To sit the final exam, you should obtain at least 30 marks during the semester.

Textbook

- Cryptography and Embedded Systems Security
- SpringerLink: `https://link.springer.com/book/10.1007/978-3-031-62205-2`
- Free version: `https://xiaoluhou.github.io/Textbook.pdf`
- Library: 6 copies, I*6K-Kryptografia, šifrovanie



Mistakes in the book

- 1 mistake – 1 mark
- At most 10 marks in total
- Mistakes include grammar errors, confusing sentences, etc.
- Report mistakes via the Teams group chat, by email, or in person.
- Only newly found errors will be awarded marks
- Errors identified during the lectures do not count
- Up-to-date version: `https://xiaoluhou.github.io/Textbook.pdf`
- Errata: `https://xiaoluhou.github.io/Errata.pdf`

Presentations

- Week 4: Presentation of Assignment 3
- Week 9: Presentation of Assignment 4
- Week 10: Presentation of Assignment 5
- Students may choose one of the following presentation options:
  - Book an online presentation slot by email at least three days in advance; it will be scheduled for the day before the tutorial (preferred option)
  - Present in person during the tutorial
    - As lectures may finish early, please arrive at least 30 minutes before the tutorial begins
  - Submit a medical certificate to the Study Department and arrange an online presentation for a later date

Course materials

https://xiaoluhou.github.io/Teaching_material/

or

https://github.com/XIAOLUHOU/Teaching_material

Note

- Some lectures might be online—please pay attention to announcements in the Teams group chat and in emails.

Datasets and analysis code for SCA

All datasets and analysis code related to SCA can be found here:

`https://github.com/XIAOLUHOU/SCA-measurements-and-analysis----Experimental-results-for-textbook/tree/main`

Extra reading materials

- Menezes, A. J., Van Oorschot, P. C., & Vanstone, S. A. (2018). Handbook of applied cryptography. CRC press.
  - Free online: `https://cacr.uwaterloo.ca/hac/`
- Stinson, Douglas R. Cryptography: theory and practice. Chapman and Hall/CRC, 2005.
  - Free online: `http://sutlib2.sut.ac.th/sut_contents/H97066.pdf`
- Lecture notes, Hardware and Embedded Systems Security, `https://creativecommons.org/licenses/by-sa/3.0/`
- Buchmann, Johannes. Introduction to cryptography. Vol. 335. New York: Springer, 2004.
- Mangard, Stefan, Elisabeth Oswald, and Thomas Popp. Power analysis attacks: Revealing the secrets of smart cards. Vol. 31. Springer Science & Business Media, 2008.
- EC council course, `https://codered.eccouncil.org/course/cryptography-and-embedded-systems-security?logged=true`

Why are we interested in cryptography?

- Cryptography is an indispensable tool for protecting information in computing systems
- It is used everywhere and by billions of people worldwide on a daily basis
  - ATM
  - Email
  - Electronic passports
  - Security tokens
  - ...

picture source: https://en.wikipedia.org/wiki/Visa_requirements_for_Slovak_citizens
https://www.slovensko.sk/_img/CMS4/Navody/eID_client_Application_User_Guide.pdf

Why are we interested in physical attacks?

- Cryptography provides algorithms that enable secure communication in theory
- In the real world, these algorithms have to be implemented on real devices:
  - Software implementations: microcontrollers
  - Hardware implementations: FPGAs
- To evaluate the security level of cryptographic implementations, it is necessary to include a physical security assessment

Targets and Attack Goals

Targets

- Credit cards
- Passports
- Key fob
- ...

Goals:

- Recovery of the secret key
- Privilege escalation
- IP theft
- ...

picture source: https://goodtimes.ca/how-many-credit-cards-is-too-many/

Different Physical Attack Methods

- Side-channel analysis attacks
  - EM/Power analysis
  - Timing analysis
  - Cache attacks
- Fault attacks
  - Optical fault injection
  - Electromagnetic fault injection
  - Clock/voltage glitch
- Hardware Trojans
- ...

picture source: https://nl.dreamstime.com/stock-foto-s-hamer-die-computer-boos-tonen-raken-image34210923

What will we cover in this course?

- Abstract algebra and number theory (week 1)
- Introduction to cryptography (week 2)
- Modern cryptography and implementations (week 3 – 4)
- Power analysis attacks, fault attacks, and countermeasures (week 5 – 11)
- Practical aspects of physical attacks (week 12)
  - Invited speaker, Dr. Jakub Breier, Senior security manager, TTControl GmbH
- Consultation session (week 12)—most exam questions will be based on examples from the lectures and will be similar to past exam questions

An important notation

- The decimal separator is denoted by ., not by ,.

$$\frac{1}{2} = 0.5$$

- We do not use the following notation:

$$\frac{1}{2} = 0,5$$

- Ten thousand is written as $10,000$
- If there are other potentially confusing notations, please let me know.

PhD Research Topics

- **Fault Attacks and Countermeasures**
  - Cryptographic implementations
  - Neural network implementations
- **Side-Channel Analysis: Attacks and Countermeasures**
  - Cryptographic implementations
  - Neural network implementations
  - AI for side-channel analysis

- You are strongly encouraged to pursue your Ph.D. studies with me
- This course provides a solid foundation and a strong starting point for PhD research
- The later modules of the course are based on recent research publications